

PRISM

A Multi-Pillar Blockchain for Security, Useful Computation and Network Participation

Whitepaper v0.1 — Draft

Abstract

Prism is a blockchain protocol designed around three complementary economic pillars: Proof of Stake, Proof of Useful Work and Proof of Useful Participation.

Rather than forcing a single consensus mechanism to represent every valuable activity within a decentralized network, Prism separates three fundamental contributions.

Proof of Stake secures the blockchain.

Proof of Useful Work converts computational resources into verifiable useful computation.

Proof of Useful Participation rewards actors who actively contribute to the growth, availability and usefulness of the network.

Prism therefore proposes a simple principle:

Proof of Stake secures. Proof of Useful Work computes. Proof of Useful Participation grows.

The protocol has a fixed maximum supply of 100,000,000 PRISM.

1. Motivation

Blockchain networks must solve several different economic problems simultaneously.

They must remain secure.

They must attract computational resources.

They must encourage users, developers and infrastructure providers to participate in the network.

Existing systems generally optimize primarily for one of these objectives.

Proof-of-Work networks economically reward computational expenditure.

Proof-of-Stake networks economically reward capital committed to securing consensus.

Prism introduces a third dimension by explicitly rewarding useful network participation while also allowing computational resources to perform verifiable productive work.

The objective is not to replace Proof of Stake with another consensus mechanism.

Instead, Prism uses Proof of Stake as its security foundation and builds two additional contribution layers around it.

2. Design Principles

Prism is built around five principles.

2.1 Security must remain simple

Consensus must remain understandable and auditable.

For this reason, Proof of Stake is responsible for block production, ordering and finality.

Proof of Useful Work and Proof of Useful Participation do not compete with validators to determine the canonical blockchain.

2.2 Computation should produce measurable value

Computational resources should ideally perform useful tasks rather than arbitrary repeated calculations.

A computation can only be rewarded when its result can be independently verified by the protocol.

2.3 Participation has economic value

A decentralized network depends on more than validators.

Nodes provide availability, users govern the protocol, infrastructure providers serve data and contributors improve the ecosystem.

Prism aims to make these contributions visible within the protocol economy.

2.4 Contributions must be verifiable

Prism should not reward unverifiable claims.

Security, computation and participation rewards must ultimately derive from cryptographically or economically verifiable actions.

2.5 The monetary supply must remain predictable

The maximum supply of PRISM is permanently capped at:

3. Protocol Architecture

Prism separates the protocol into three contribution layers.

3.1 Proof of Stake — Security Layer

Proof of Stake forms the consensus foundation of Prism.

Validators lock PRISM as economic collateral.

Validators may:

- propose blocks;
- attest proposed blocks;
- participate in finality;
- verify protocol rules;
- validate PoUW and PoUP receipts.

Validator voting power is derived from stake according to protocol rules.

A block becomes final after receiving the required supermajority of validator stake.

Validators who intentionally violate consensus rules may lose part of their stake through slashing.

Examples include:

- double signing;
- conflicting attestations;
- deliberate validation of invalid state transitions.

Proof of Stake therefore protects the integrity and ordering of the Prism blockchain.

4. Proof of Useful Work

Proof of Useful Work creates a decentralized market for verifiable computation.

A computation request is represented by a Prism Job.

A Job contains information such as:

- job identifier;
- computation type;
- input commitment;
- verification rules;

- reward;
- deadline.

Workers may accept compatible jobs and execute them outside the consensus-critical block production path.

After computation, the worker submits:

- the result;
- a commitment to the result;
- the required proof or verification data.

Validators then verify the submitted receipt according to the rules defined for the job type.

Only verified work becomes eligible for PoUW rewards.

The first generations of Prism should prioritize deterministic workloads whose results can be cheaply checked.

Future versions may integrate technologies such as:

- zero-knowledge proofs;
- verifiable computation;
- distributed computation;
- cryptographic challenge systems.

The long-term goal of PoUW is to allow blockchain economic incentives to finance useful computational activity.

5. Proof of Useful Participation

Proof of Useful Participation represents the third pillar of Prism.

PoUP rewards measurable actions that improve the network without necessarily requiring large amounts of capital or computational power.

Eligible participation may include:

- node availability;
- successful network challenges;
- serving blockchain data;
- protocol governance;
- infrastructure services;
- ecosystem contributions supported by verifiable mechanisms.

Each eligible activity contributes to a Participation Score.

Rewards are distributed according to verified participation rather than raw activity volume.

The protocol must specifically resist artificial participation.

Simple metrics such as transaction count are therefore unsuitable because they can encourage spam.

PoUP should instead use mechanisms including:

- diminishing returns;
- reward caps;
- randomized challenges;
- reputation or history;
- economic bonds where appropriate;
- Sybil resistance.

The long-term objective is to create a network in which useful participation has explicit economic value.

6. Prism Blocks

A Prism block contains the traditional blockchain state transition data as well as commitments to useful work and participation.

A conceptual Prism block contains:

Header

- block height;
- timestamp;
- previous block hash;
- proposer identity;
- state root.

Transactions

Normal PRISM state transitions.

PoUW receipts

Records of useful computation accepted by the network.

PoUP receipts

Records of verified network participation.

Validator attestations

Consensus signatures confirming the validity and finality of the block.

The block header may therefore commit to several Merkle roots:

- transaction root;
- useful work root;
- participation root;
- global state root.

This structure keeps each subsystem independently auditable.

7. Epochs and Reward Distribution

Prism distributes rewards over protocol epochs.

Each epoch independently accounts for contributions from:

- Proof of Stake;
- Proof of Useful Work;
- Proof of Useful Participation.

The initial monetary model reserves 60,000,000 PRISM for network incentives.

The pools are divided equally:

20,000,000 PRISM — Proof of Stake

20,000,000 PRISM — Proof of Useful Work

20,000,000 PRISM — Proof of Useful Participation

Separating the reward pools prevents one category of activity from consuming rewards intended for another pillar.

Each pillar may ultimately use a different emission curve.

Proof of Stake rewards may depend on validator performance and stake.

Proof of Useful Work rewards may depend on verified computational work.

Proof of Useful Participation rewards may depend on verified participation scores.

8. Supply

The maximum supply of the Prism network is:

100,000,000 PRISM

The preliminary allocation is:

- 20M — Proof of Stake rewards;
- 20M — Proof of Useful Work rewards;
- 20M — Proof of Useful Participation rewards;
- 15M — Ecosystem;
- 10M — Treasury;
- 10M — Team;
- 5M — Liquidity and network launch.

No mechanism should allow the total supply to exceed the maximum supply.

9. Security Model

Prism separates security concerns across its three pillars.

Consensus attacks

Proof of Stake attacks are mitigated through economic collateral, validator attestations and slashing.

Fraudulent useful work

Workers must provide verification material for submitted computations.

Invalid work receives no reward and may result in loss of bonded collateral.

Participation farming

PoUP rewards must use capped and Sybil-resistant scoring mechanisms rather than unlimited activity counters.

Network liveness

PoUW and PoUP are deliberately separated from block finality.

The blockchain should remain capable of producing blocks even when no useful computation jobs or participation receipts are available.

This separation is essential to Prism's security model.

10. Governance

Prism governance may eventually control protocol parameters including:

- reward curves;
- accepted PoUW job classes;

- participation scoring;
- validator parameters;
- treasury spending;
- ecosystem incentives.

Critical monetary and security constraints should remain protected by strict governance rules.

Governance should not be capable of silently bypassing the maximum PRISM supply.

11. Development Roadmap

Phase 1 — Core Blockchain

Genesis block, transactions, state management and basic chain validation.

Phase 2 — Proof of Stake

Validator registry, staking, proposer selection and block attestations.

Phase 3 — Proof of Useful Work

Job marketplace, worker protocol, deterministic computation and verifiable PoUW receipts.

Phase 4 — Proof of Useful Participation

Participation challenges, scoring engine and PoUP reward distribution.

Phase 5 — Integrated Prism Network

Unified epochs, reward engine, validator consensus and public testnet.

Phase 6 — Advanced Verification

Research and implementation of zero-knowledge or other verifiable-computation mechanisms for larger classes of useful work.

12. Vision

Prism is based on the idea that decentralized networks rely on several forms of contribution.

Capital provides economic security.

Computational resources provide processing capability.

Humans and infrastructure provide participation.

A blockchain economy does not necessarily need to collapse all three into a single mechanism.

Prism instead recognizes each contribution explicitly.

Proof of Stake secures.

Proof of Useful Work computes.

Proof of Useful Participation grows.

Together, they form Prism.